

Lab Report: Single Sideband (SSB) Modulation and Demodulation

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1 Objective

To study the generation and demodulation of a **Single Sideband (SSB) signal** using an analog/digital trainer kit (EMONA Telecoms Trainer 101) and verify the results on a digital storage oscilloscope (Keysight DSO1014A).

2 Basic Theory

2.1 Amplitude Modulation (AM)

In amplitude modulation, a carrier signal $c(t) = A_c \cos(2\pi f_c t)$ is multiplied with a message signal $m(t)$, producing

$$\text{AM}(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t).$$

This produces a double sideband (DSB) signal with two symmetric sidebands around the carrier.

2.2 Double Sideband Suppressed Carrier (DSB-SC)

Removing the carrier results in

$$\text{DSB}(t) = m(t) \cos(2\pi f_c t),$$

which has both upper and lower sidebands carrying the same information.

2.3 Single Sideband (SSB)

By suppressing one sideband using the phase-shift (Hilbert transform) method or via filtering, we get

$$\text{SSB}(t) = m(t) \cos(2\pi f_c t) \pm \hat{m}(t) \sin(2\pi f_c t),$$

where $\hat{m}(t)$ is the Hilbert transform of $m(t)$. This reduces the required transmission bandwidth by half compared to AM.

2.4 Demodulation

At the receiver, coherent detection with a synchronized carrier is used:

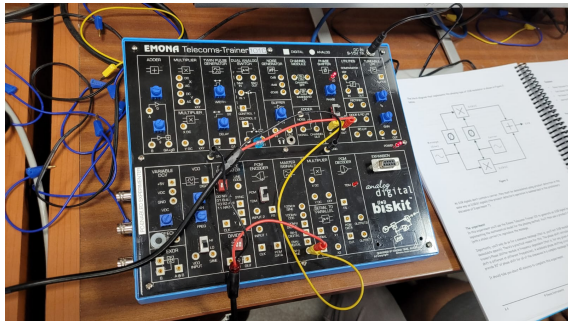
$$\hat{m}(t) = \text{SSB}(t) \cdot \cos(2\pi f_c t),$$

followed by a low-pass filter to recover the message.

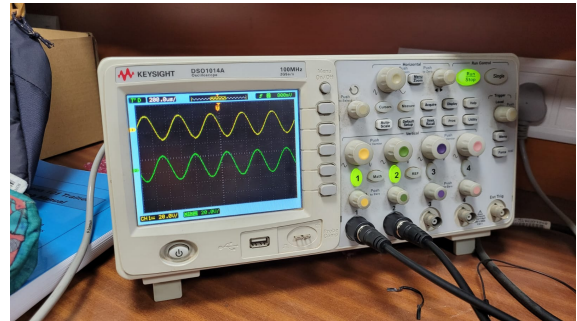
3 Experimental Setup

The setup was implemented on the **EMONA Telecoms Trainer 101** with the following steps:

1. **Message Signal Generation and Phase Shifting:** Generate a low-frequency sinusoidal message signal and obtain its quadrature (90° phase-shifted) version using a Hilbert transform module.



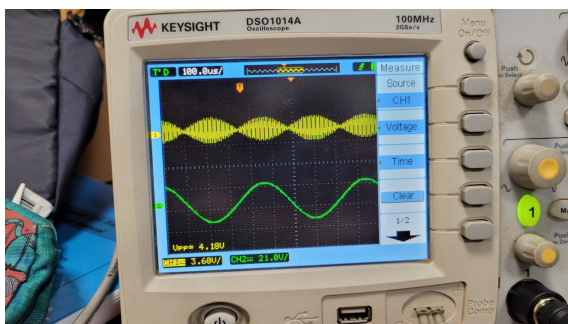
(a) Message signal generation setup



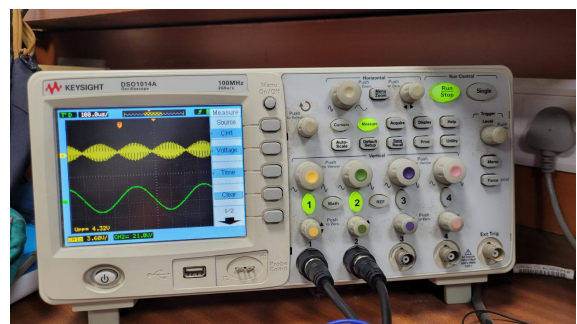
(b) DSO output: message signal and its 90° phase-shifted version

Figure 1: Step 1: Message signal generation and phase shifting

2. **Multiplication with Carrier Signals:** Multiply the message signal with a cosine carrier, while the Hilbert-transformed signal is multiplied with a sine carrier of the same frequency.



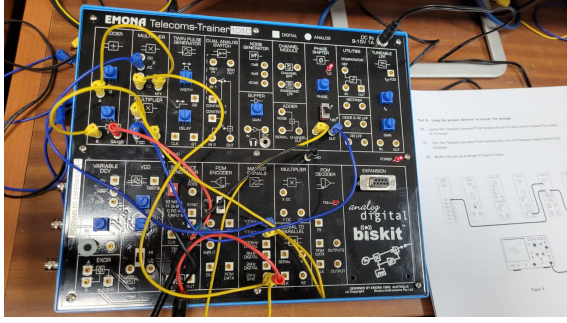
(a) DSO output: $m(t) \times \cos(2\pi f_c t)$



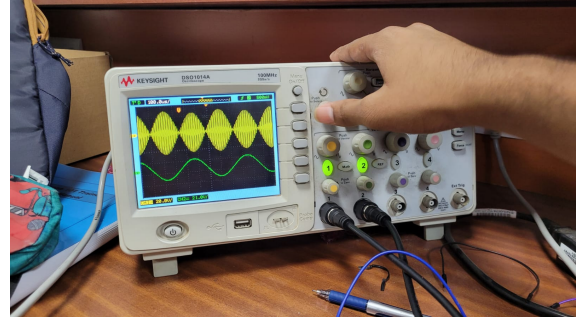
(b) DSO output: $\hat{m}(t) \times \sin(2\pi f_c t)$

Figure 2: Step 2: Multiplication with carrier signals

3. **SSB Signal Formation:** Combine the two multiplied signals to form the Single Sideband signal, effectively suppressing one of the sidebands present in conventional AM.

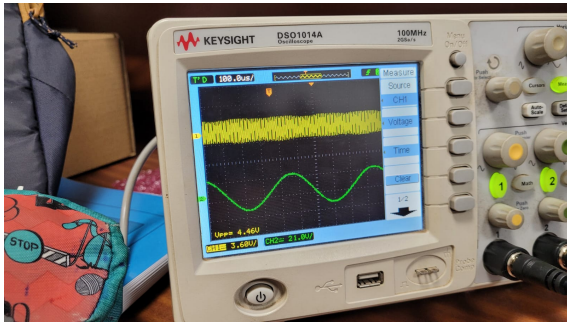


(a) SSB formation setup

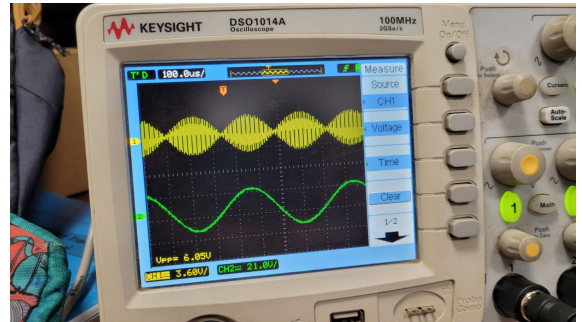


(b) DSO output: DSB-SC reference signal

Figure 3: Step 3a: SSB formation setup and DSB-SC reference



(a) SSB output after amplitude adjustments



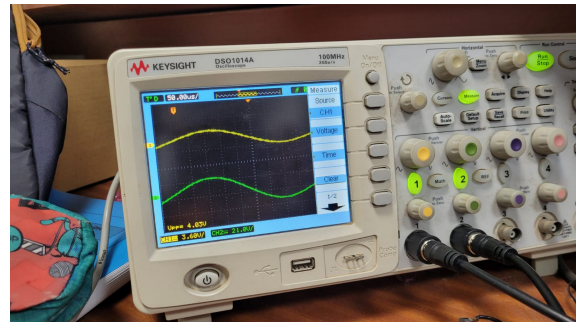
(b) DSO output: final SSB modulated waveform

Figure 4: Step 3b: SSB signal formation results

4. **Demodulation and Signal Recovery:** Apply a product detector (coherent detection) to demodulate the SSB signal and recover the original message signal.



(a) Complete setup including demodulation circuit



(b) DSO output: original message signal and recovered signal comparison

Figure 5: Step 4: Demodulation and signal recovery

4 Results

1. The message signal was observed as a low-frequency sinusoid (Figure 1b).
2. The Hilbert transformed signal showed a clear 90° phase shift (Figure 1b).

3. Multiplication with carrier signals produced two intermediate signals of different phases (Figures 2a and 2b).
4. Adding them produced the SSB modulated waveform, with clear suppression of one sideband compared to DSB (Figures 3b, 4a, and 4b).
5. On demodulation using a product detector, the original message waveform was recovered (Figure 5b).